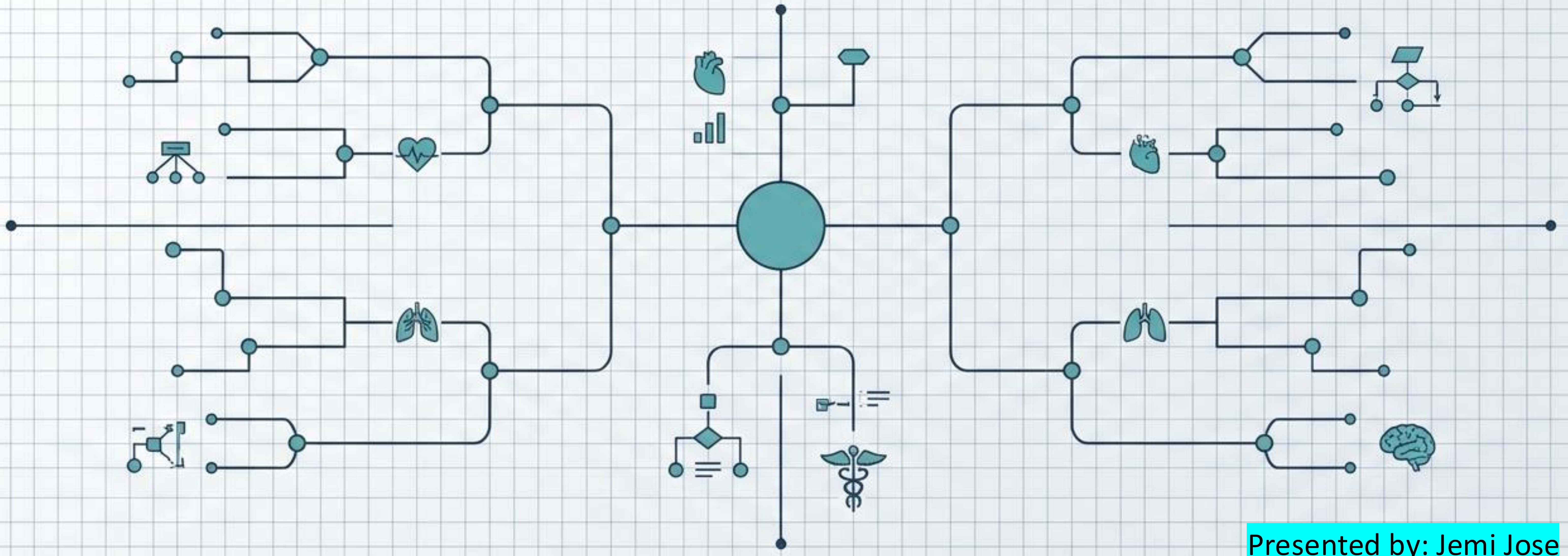
AI DOCTOR

An Intelligent Disease Prediction and Triage System



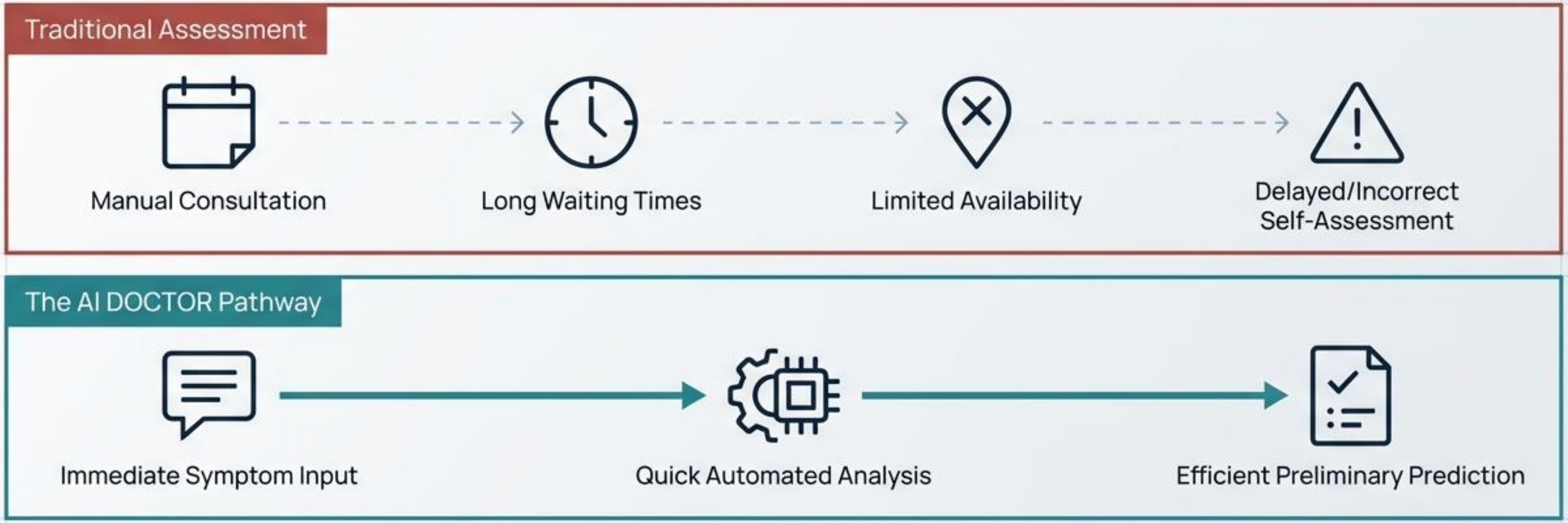
Presented by: Jemi Jose

A technical case study detailing a full-stack machine learning web application designed to bridge the gap between patient symptoms and preliminary health awareness.

The Healthcare Waiting Journey

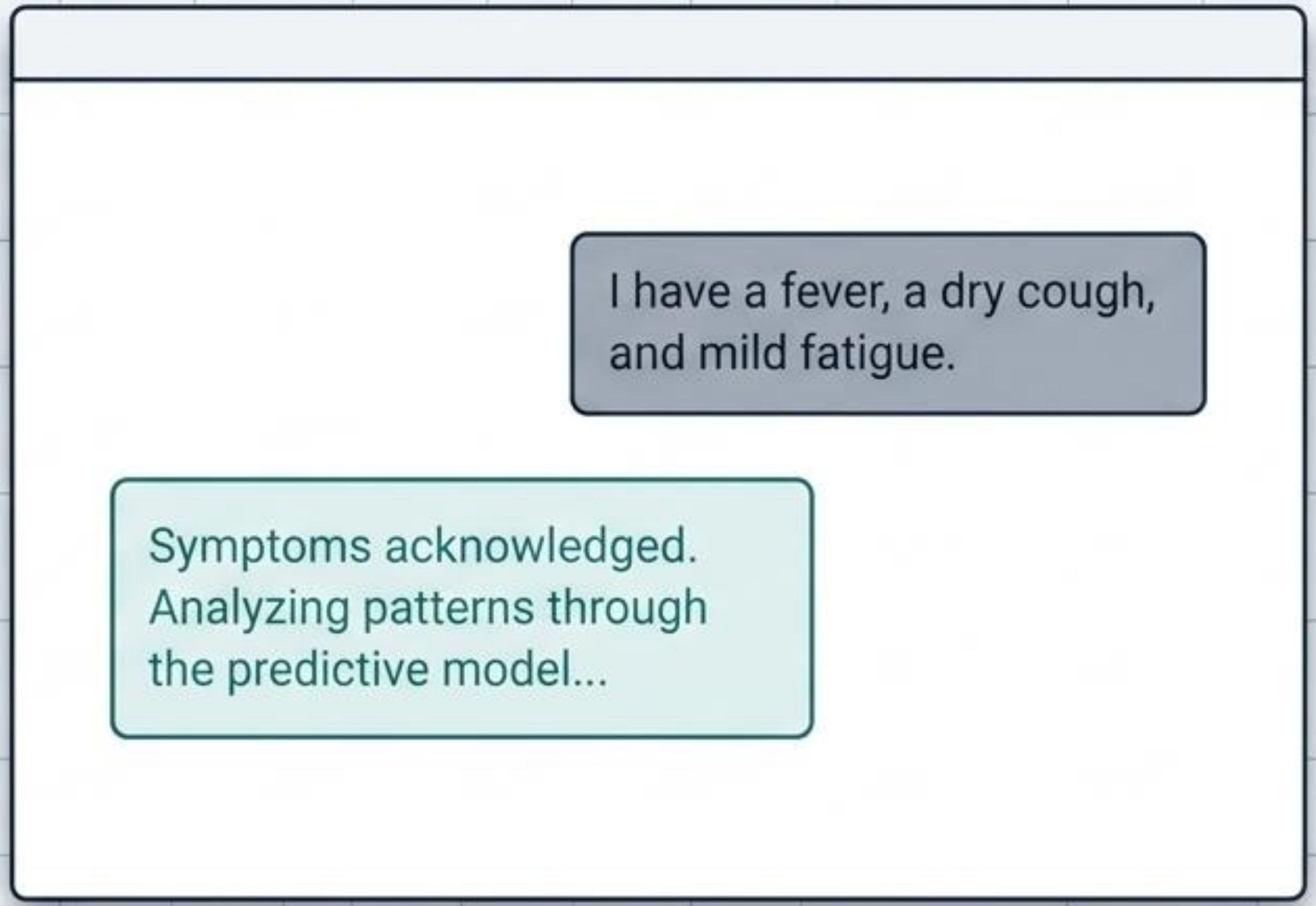
Individuals often lack the medical knowledge to accurately interpret their symptoms, requiring a system that efficiently processes inputs and guides users toward appropriate care.

Friction Map



Automated Preliminary Analysis at Scale

AI DOCTOR is a full-stack web application powered by machine learning. It provides non-technical users with an interactive, chat-based interface to input symptoms and instantly receive the most probable disease prediction alongside a confidence score.



Evaluating Assessment Methodologies

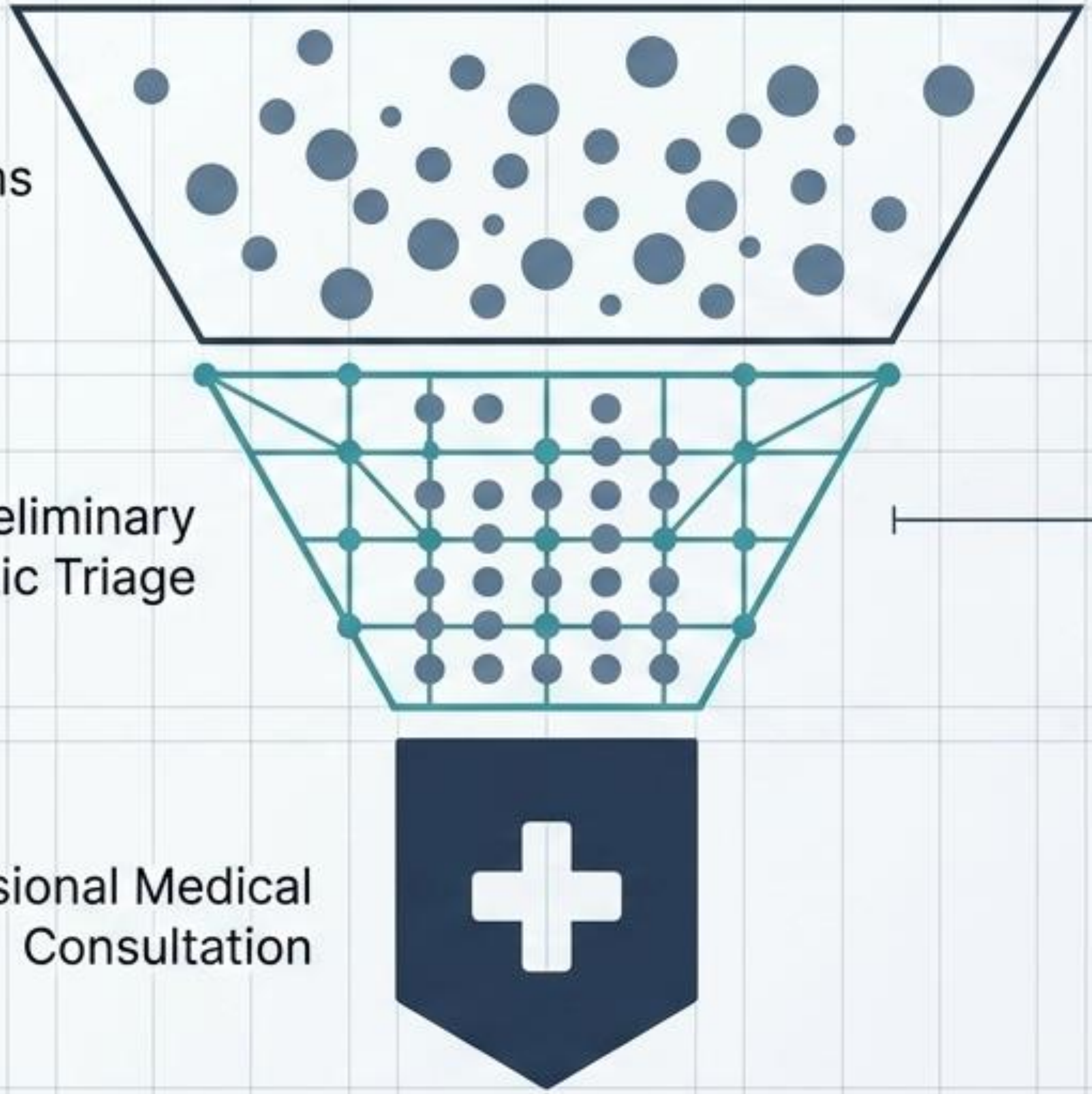
Assessment Dimension	Traditional Self-Assessment	AI DOCTOR
Speed	Delayed by wait times	✓ Real-time generation
Accessibility	Geographically and temporally limited	✓ 24/7 web access
Data Complexity Handling	Prone to human misinterpretation	✓ Analyzes complex, high-dimensional symptom patterns
Output Reliability	Anecdotal and uncertain	✓ Algorithmic prediction with an explicit % confidence score

The Assistive Triage Layer

Patient Symptoms

AI DOCTOR: Preliminary Diagnostic Triage

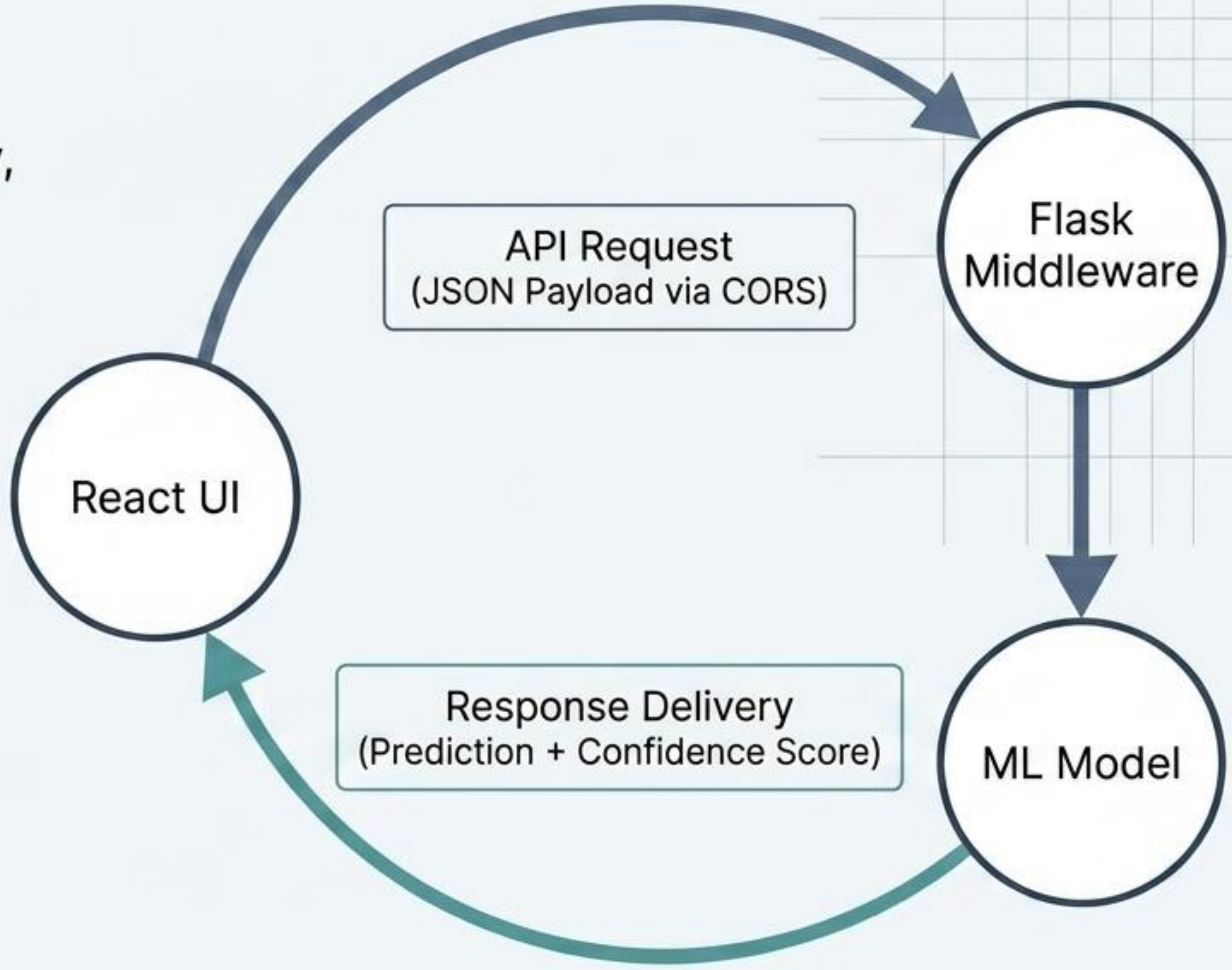
Professional Medical Consultation



While it provides early-stage health awareness and decision support, this system acts strictly as an assistive tool. It maps patterns hidden in large datasets but is not a replacement for professional clinical diagnosis.

The 3-Tier API Request Cycle

A modular client-server design ensures scalability and maintainability, separating the interactive user experience from heavy predictive computation.



Diagnostic Breakdown of the Tech Stack

Frontend (The Interface)

Technologies: React.js, CSS

Functional Role: Drives the interactive, non-technical chat interface and renders the UI.

Backend (The Router)

Technologies: Flask (Python), Flask-CORS

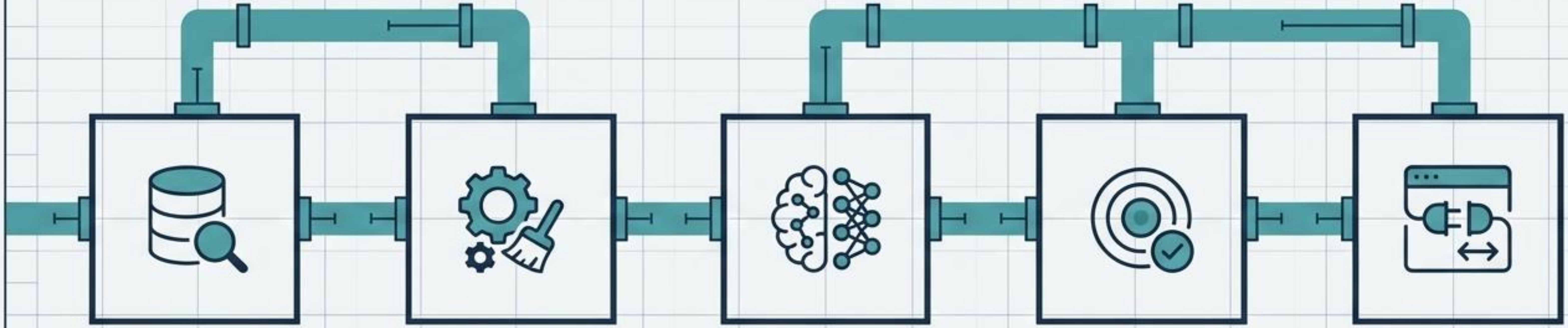
Functional Role: Manages cross-origin communication and serves as the API bridge between user input and the model.

Machine Learning (The Engine)

Technologies: Scikit-learn, Random Forest Classifier

Functional Role: Analyzes structured symptom patterns to generate probable disease outcomes.

The Machine Learning Data Refinery



Step 1: Collection

Sourced from Kaggle Symptom-Disease Dataset.

Step 2: Preprocessing

Handling missing values, binary symptom encoding, train/test splitting.

Step 3: Training

Fitting the Random Forest Classifier.

Step 4: Evaluation

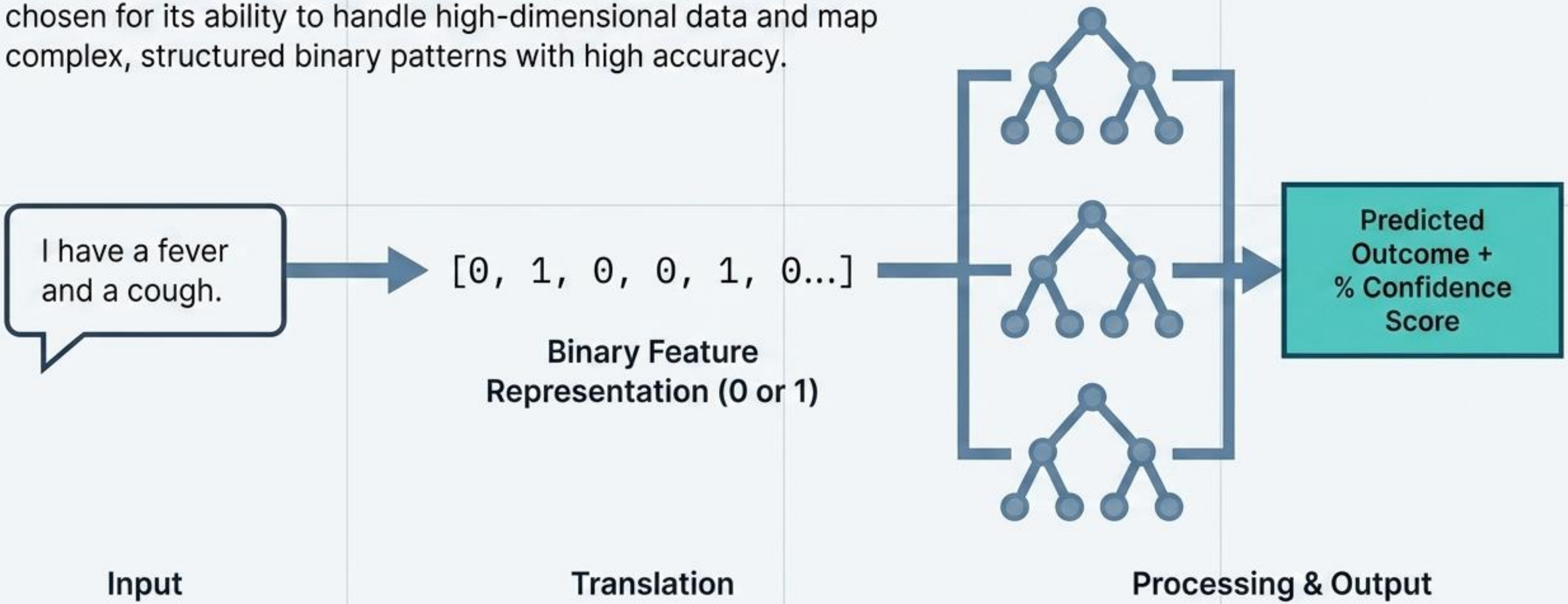
Verifying high accuracy via scoring and cross-validation.

Step 5: Integration

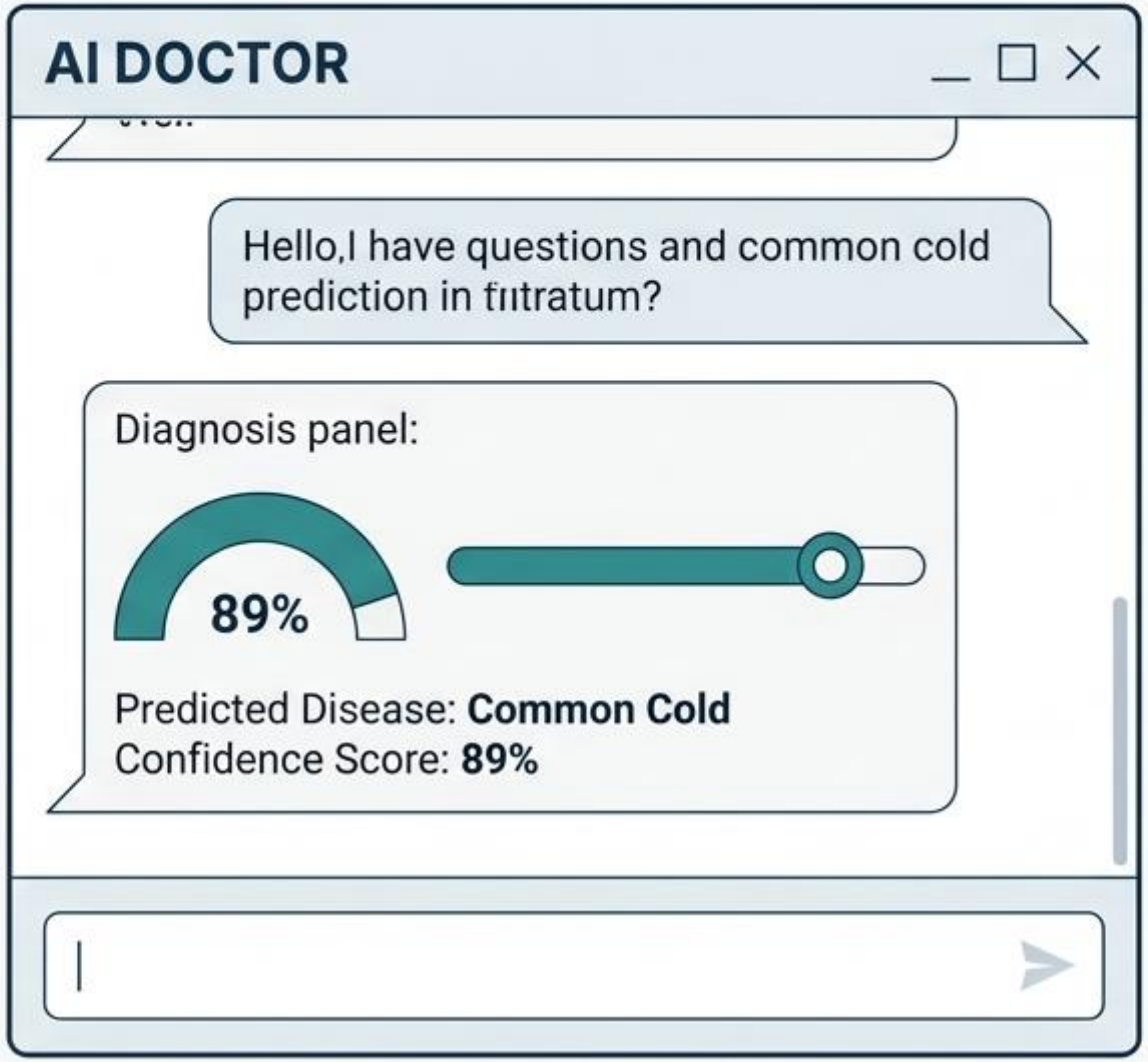
Deploying the trained model into the Flask API.

From Natural Symptom to Algorithmic Prediction

The system utilizes a Random Forest algorithm specifically chosen for its ability to handle high-dimensional data and map complex, structured binary patterns with high accuracy.



High-Accuracy Real-Time Inference



Real-Time Processing

Instant prediction generation.

Proven Accuracy

High performance driven by the structured Kaggle dataset and Random Forest effectiveness.

Accessible UX

Chat-based formatting removes technical barriers for end-users.

System Boundaries and Clinical Realities

Boundary Matrix

Current Scope (Capabilities)	Clinical Reality (Limitations)
• Highly accurate on structured datasets • Excels at predefined symptom-disease mappings • Fast binary pattern recognition	• Cannot process unstructured or ambiguous text input • Struggles with overlapping symptom complexities not captured in the training data • Cannot issue clinical diagnoses

Medical reality is inherently complex. These boundaries define the system as an early-insight tool, strictly bounded by its predefined mappings.

Future Evolution of the Platform

Conversational AI

Integrating Natural Language Processing (NLP) to understand complex, unstructured user inputs.

Voice Accessibility

Adding voice-based input functionality to lower usage barriers.

Enhanced Generalization

Training on more diverse, real-world medical datasets to capture overlapping symptom complexities.

Global Scalability

Deploying the architecture to cloud platforms for wider public access.

Bridging the Healthcare Information Gap

AI DOCTOR proves that integrating machine learning with modern web APIs can create fast, interactive, and highly accessible decision support systems. By providing early insights through a seamless interface, we demonstrate the vital role of artificial intelligence in the future of healthcare accessibility.